

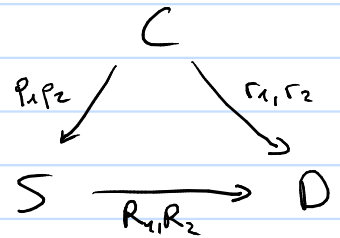
Smoking & Cornfield's inequality

Let S be the population of smokers, D the population of people with lung cancer and C the population with a smoking gene, the only cause for $P(D|S) > P(D|\neg S)$. We will show with Cornfield's inequality that it is unrealistic for such a gene to explain how much bigger $P(D|S)$ is compared to $P(D|\neg S)$.

Let

$$\begin{aligned} r_1 &= P(D|C), & p_1 &= P(C|S), & R_1 &= P(D|S) \\ r_2 &= P(D|\neg C), & p_2 &= P(C|\neg S), & R_2 &= P(D|\neg S). \end{aligned}$$

Since C causes S , we have



Assumption 1: $r_1 > r_2$

If we assume that C fully explains the fact that $R_1 > R_2$, we also need

Assumption 2: $r_1 = P(D|C) = P(D|C, S) = P(D|C, \neg S)$
 $r_2 = P(D|\neg C) = P(D|\neg C, S) = P(D|\neg C, \neg S)$

What we want to explain with C is

Fact 1: $R_1 > R_2$

Now,

$$\begin{aligned} R_1 &= P(D|S) \\ &= P(D|S, C)P(C|S) + P(D|S, \neg C)P(\neg C|S) \\ &= P(D|C)P(C|S) + P(D|\neg C)P(\neg C|S) \\ &= r_1 p_1 + r_2 (1 - p_1) \\ &= p_1 (r_1 - r_2) + r_2, \end{aligned} \quad \begin{array}{l} \text{[Total probability]} \\ \text{[Ass. 2]} \end{array}$$

and similarly,

$$R_2 = P(D|\neg S) = p_2 (r_1 - r_2) + r_2.$$

Hence,

$$\frac{R_1}{R_2} = \frac{p_1 (r_1 - r_2) + r_2}{p_2 (r_1 - r_2) + r_2} = \frac{p_1 + x}{p_2 + x},$$

where

$$x = \frac{r_2}{r_1 - r_2}.$$

Therefore

$$\begin{aligned} \frac{R_1}{R_2} (p_2 + x) &= p_1 + x \\ \Leftrightarrow \frac{R_1}{R_2} &= \frac{p_1}{p_2} + \frac{x}{p_2} \left(1 - \frac{R_1}{R_2}\right) \\ \Rightarrow \frac{R_1}{R_2} &< \frac{p_1}{p_2} \end{aligned}$$

[Fact 1 + Ass. 1]

From observational data, we have $R_1/R_2 > 9$, so $p_1 > 9p_2$. In particular, $p_2 < \frac{1}{9} < 12\%$, and the larger p_2 is, the closer p_1 has to be to 100%. For example $p_2 = 11\%$ then $p_1 > 99\%$. It is very unrealistic that a gene could have such a big impact on the complex decision whether to smoke or not. So there must be at least some causal effect of S on D .